Lab 14

Introduction to Micro Blaze processor

Spring 2025

Submitted by: Syeda afia shah

Registration No: 22pwsce2115

Class Section: B

"On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work."

Student Signature: ___ Uswa _____

Submitted to:

Engr. Faheem Jan

6/20/2025

Department of Computer Systems Engineering

University of Engineering and Technology, Peshawar

Objective

To learn how to use Xilinx MicroBlaze Microcontroller

Introduction

MicroBlaze is Xilinx's 32-bit RISC soft processor core, optimized for embedded applications on Xilinx devices. The MicroBlaze processor is easy to use and delivers the flexibility to select the combination of peripherals, memory, and interfaces as needed. The MicroBlaze soft processor core is included with the Xilinx software tools.

Steps to follow:

1. Import the MicroBlaze IP core in the project design

2. Write Verilog code and instantiate the MicroBlaze processor and synthesize the design 3. Create New application using Xilinx SDK and generate .elf (executable and linkable file) 4. After Synthesis export the BMM file using the following command:

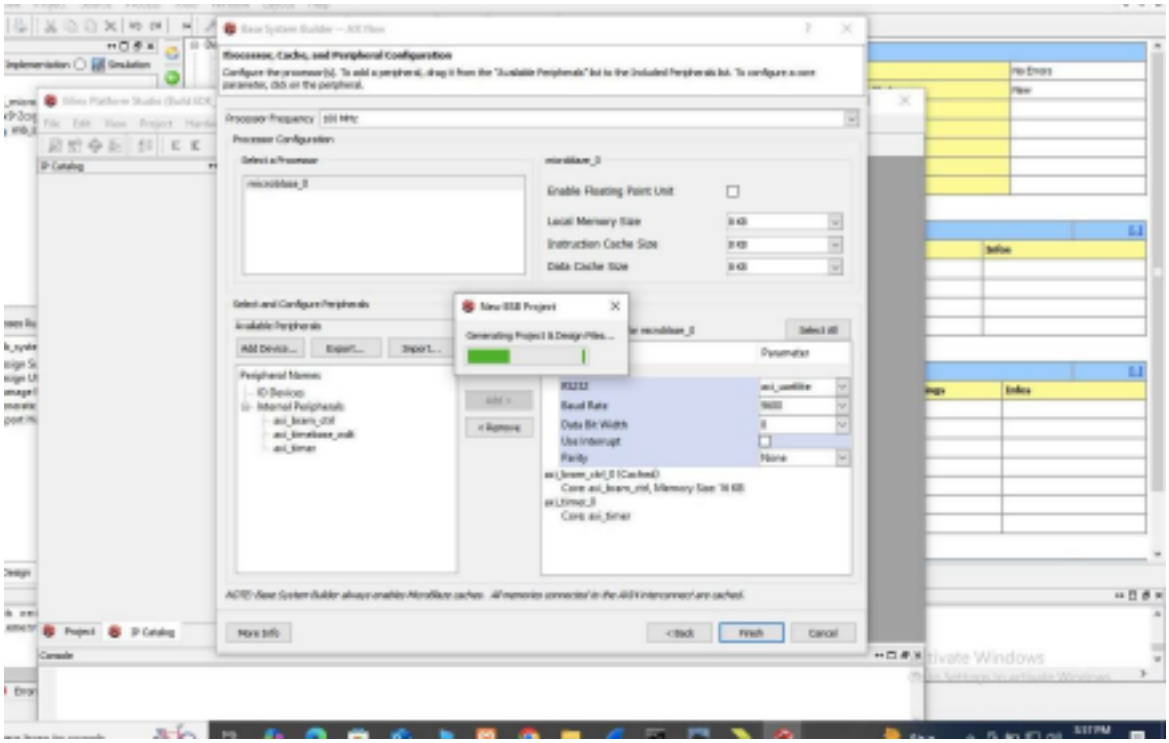
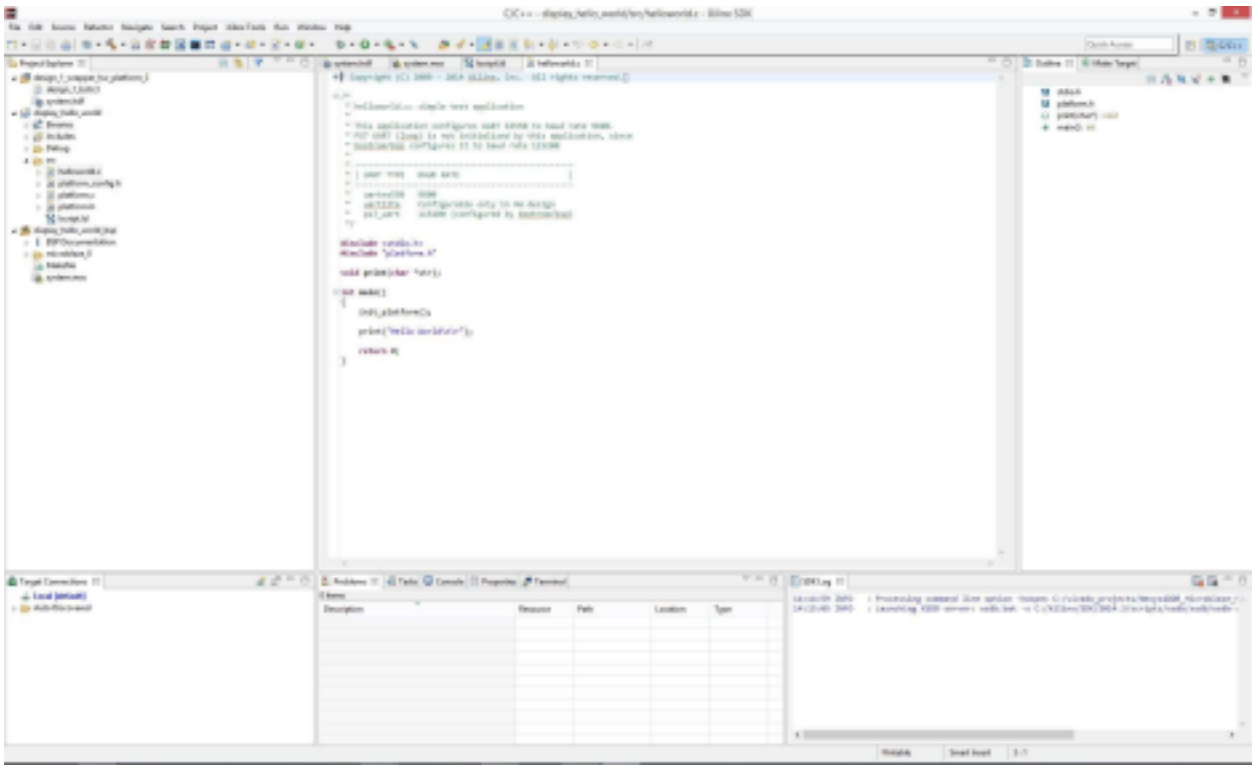
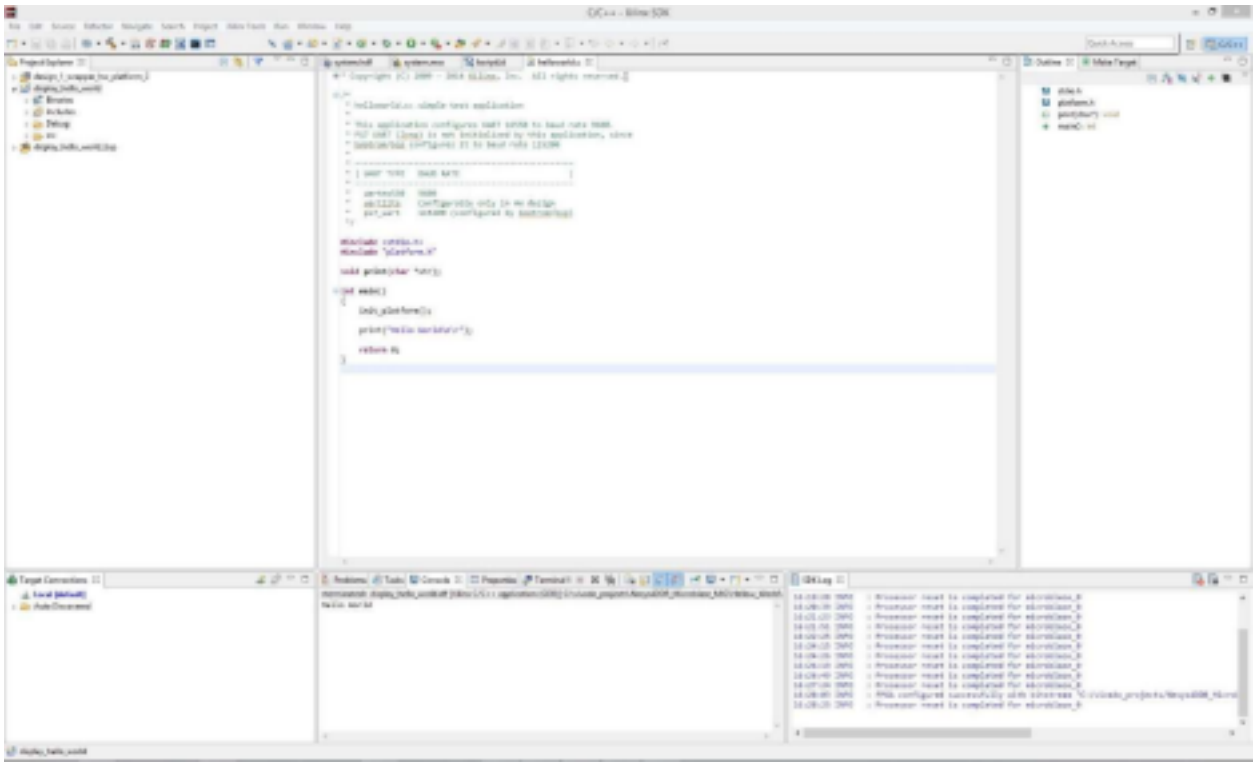
source ipcore_dir/microblaze_mcs_setup.tcl

5. Implement the design and generate bit stream

6. Merge the .elf file with bit file using the following command:

microblaze_mcs_data2mem sdk/HelloWorld/Debug/HelloWorld.elf

7. Create bin file from bit using the following command:

promgen -w -p bin -u 0x0 helloworld.bit -spi -o helloworld

PERFORMANCE: